Powered by Akumen AI : CodeRonin

GEOPULSE : INDOOR OUTDOOR POSITIONING SYSTEM

Presented by Team HackHers

THE PROBLEMS

ISSUE

The Problem: Traditional GPS fails completely indoors. Visually impaired individuals lack reliable, non-visual navigation systems for complex indoor environments.

THE AFFECTED

Visually impaired people navigating high-traffic, dynamic spaces like hospitals, university campuses, and transit hubs.

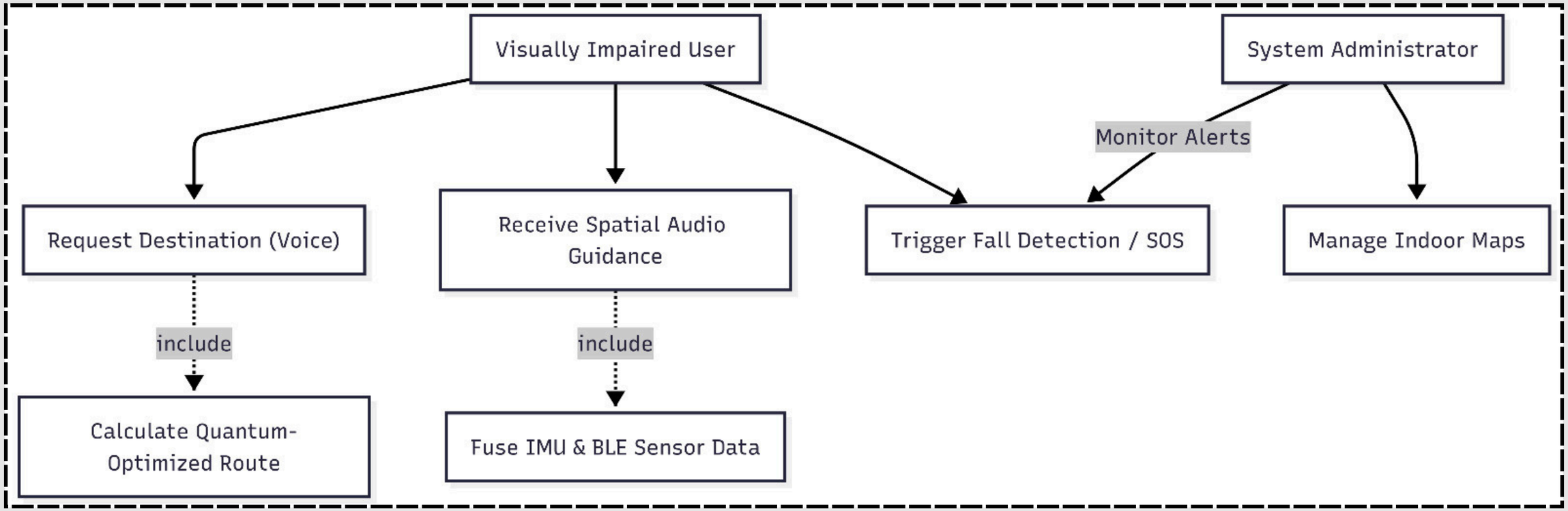
EXISTING CHALLENGES

Physical hardware beacons are expensive to deploy. Furthermore, standard navigation apps rely on visual maps (screens), which are entirely useless for this demographic.

PROPOSED SOLUTOIUON

- Core Concept: A dual-layer software system. First, we simulate an indoor beacon network using a cluster of stationary smartphones broadcasting BLE signals
- Solving Challenges: Instead of a visual map, we translate X/Y coordinates into an intuitive 3D Spatial Audio interface that guides the user via voice and sound.
- Unique Aspects: Zero-Hardware Dependency: Uses existing smartphone infrastructure.
- Quantum Routing: Integrates IBM Qiskit to instantly calculate the absolute shortest, obstacle-free path—bypassing classical NP-Hard routing bottlenecks

Our solution is uniquely architected as a Dual-Purpose Engine: it serves as a high-fidelity navigation tool for general users while providing specialized spatial-audio and safety protocols for visually impaired users.



UseCase Diagram

TECHNICAL ARCHITECTURE

Hardware Layer (Data Ingestion):

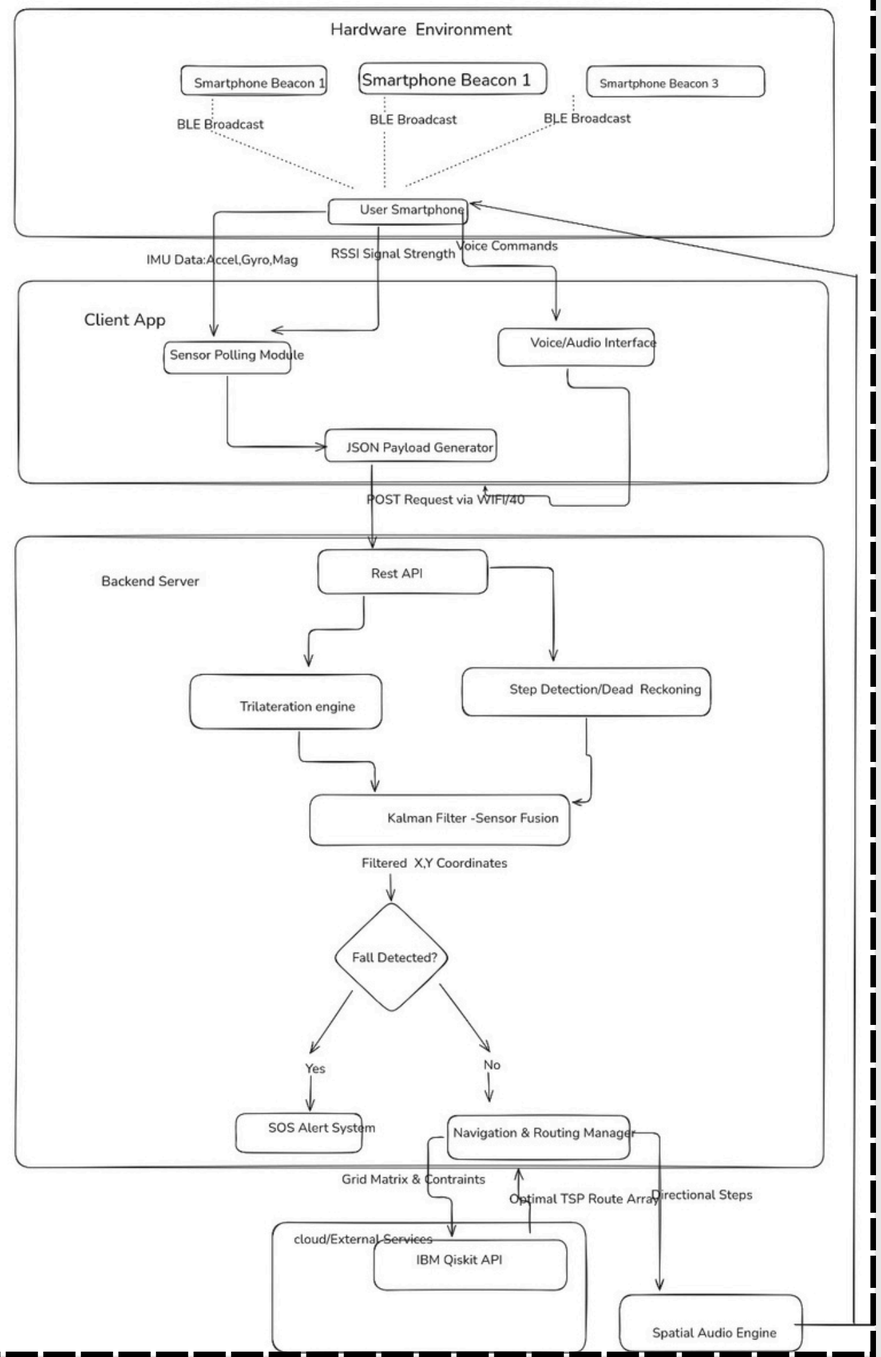
The client app continuously polls BLE signal strength (RSSI) from the stationary phones while simultaneously reading the user's IMU data (accelerometer/gyroscope).

Backend Brain (Python/Flask)

Sensor Fusion Engine: We don't just trust raw, bouncy Bluetooth signals. We use algorithms like the Kalman Filter to fuse BLE data with the IMU step-detection for pinpoint accuracy

Zero-Infrastructure Deployment

Our system eliminates the need for expensive proprietary hardware by repurposing standard smartphones into a high-density, simulated BLE beacon network .



Multimodal Sensor Fusion

To achieve sub-meter accuracy, we implement a Kalman Filter-based fusion engine that synchronizes intermittent BLE signals with continuous IMU Pedestrian Dead Reckoning (PDR)

Architecture Diagram

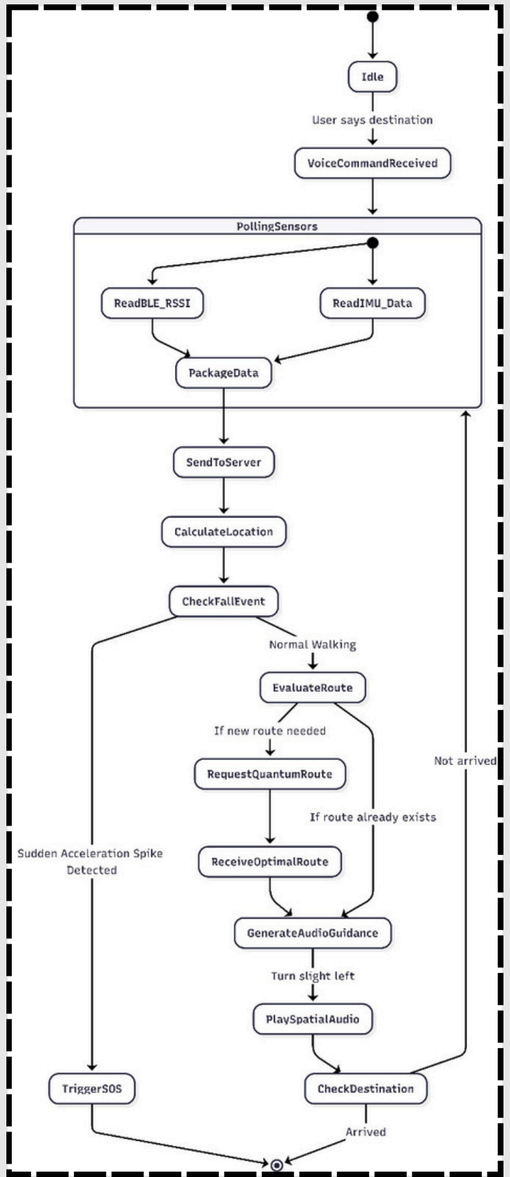
QUANTUM-OPTIMIZED PATHFINDING

- **Computational Efficiency:** Formulates building floor plans into QUBO matrices to bypass classical NP-Hard routing bottlenecks .
- **Parallel Path Analysis:** Leverages IBM Qiskit to evaluate all possible navigation routes simultaneously via quantum superposition
- **Real-Time Optimization:** Delivers the mathematically shortest, obstacle-free path in under 500ms for seamless re-routing
- **Scalable Architecture:** Ensures lag-free navigation across high-density environments like transit hubs or hospitals.

REAL-TIME NAVIGATION & DECISION LOGIC

Sensor Fusion: PEOPLEx: PEdestrian Opportunistic Positioning LEveraging IMU, UWB, BLE and WiFi.

Asynchronous Sensor Polling: To minimize latency, the client app captures Bluetooth RSSI and IMU data in parallel threads before packaging the payload for the backend



Dynamic Re-Routing: If the user deviates from the path, the system automatically triggers a new Quantum-Optimized route calculation without requiring manual intervention

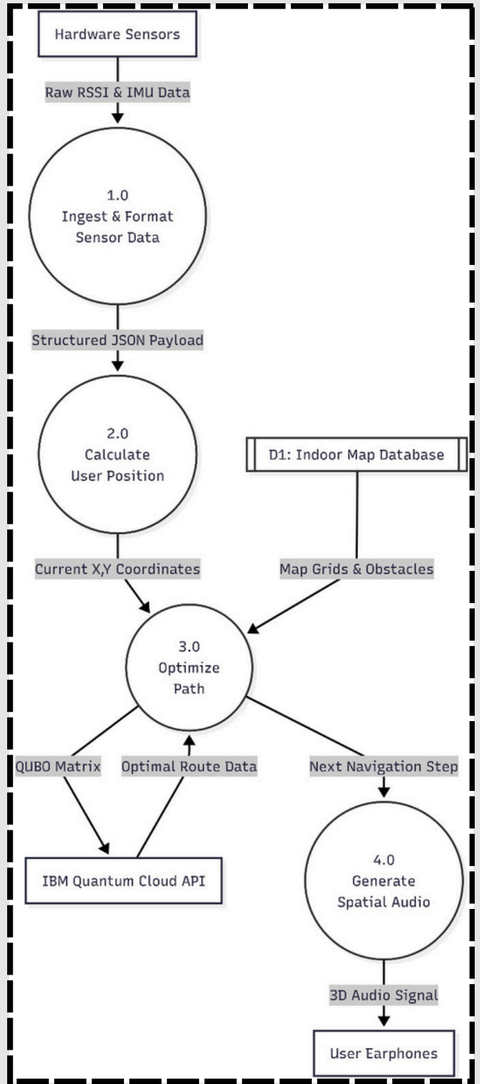
Fail-Safe Destination Tracking: The loop only terminates once the "Arrived" state is confirmed by the sensor fusion engine, ensuring the user is never left without guidance .

Activity Diagram

FEASIBILITY AND IMPLEMENTATION

Risk and Mitigation

- Risk: BLE signals bounce off walls and people, creating noisy data.
-Solution: Pedestrian Dead Reckoning (PDR). By utilizing the IMU accelerometer to track physical steps, we smooth out the signal noise and maintain accuracy even if a beacon temporarily fails



DataFlow(L-1) Diagram

Scalability

The system can be deployed in any existing building within minutes using standard mobile devices, without drilling into walls or installing expensive proprietary beacons.

RESEARCH AND REFERENCES

- Sensor Fusion: PEOPLEx: PEdestrian Opportunistic Positioning LEveraging IMU, UWB, BLE and WiFi
- Quantum Algorithms: A High-Performance Digital QUBO Solver for Quantum-Inspired Optimization
- Accessibility Interface: An Audio Augmented Reality Navigation System for Blind and Visually Impaired People
- Software Specification Requirement (SRS) : View SRS

THANKYOU

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